Dr. Aftab Alam

Curriculum Vitae - Full Academic Record

College of Medicine and Health Sciences, United Arab Emirates University, Al Ain, UAE | aftabal@uaeu.ac.ae | +917 55 599 6324
LinkedIn | Google Scholar | ResearchGate | GitHub | Orcid | Scopus | Academic Profile

CAREER OBJECTIVE

Bioinformatician with expertise in multi-omics data integration, systems biology, and precision medicine, seeking to contribute to research initiatives that translate large-scale biological data into therapeutic and diagnostic advances.

POST-DOCTORAL TRAINING

- **2023 – Present** Postdoctoral Researcher, **United Arab Emirates University** (UAEU), Al-Ain, UAE
- **2022 – 2023** Postdoctoral Researcher, **Memorial Sloan Kettering Cancer Center (MSKCC)**, NYC, USA

RESEARCH INTERESTS

- Multi-omics Data Integration
- Drug Design and Development
- Immunoinformatics
- Next-gen data and metagenomic/microbiome analyses
- Structure-Based Bioinformatics
- Personalized Medicine
- Network and Systems Biology
- Big Data Analyses

RESEARCH EXPERIENCE

Over seven years of bioinformatics experience, specializing in multi-omics data analysis across transcriptomics, proteomics, metabolomics, and microbiomics, with expertise in biological networks and systems biology, drug design and development, and personalized medicine.

TEACHING EXPERIENCE

- **PhD Teaching Assistant:** Supported teaching in Bioinformatics, *Advanced Bioinformatics Tools, Systems Biology and Applications, Microarray Data Analysis, and Structural Bioinformatics*.
- **Postdoctoral Teaching Experience (UAEU):** Contributed to the Advanced Immunology (BMSC 713; CRN 25132) course, covering *Systems Immunology, data preprocessing and integration, bioinformatics tools and pipelines, PPI network analysis, and immune infiltration analysis.*

PROFESSIONAL SKILLS

- Multi-Omics Data Integration and Systems Biology
- Machine Learning and AI in Bioinformatics
- Metagenomics and Microbiome Analysis
- Scripting experience with R/Python
- Bioinformatics education and mentoring
- Manuscript drafting and grant writing
- Structural Bioinformatics
- Project Management
- Interpersonal and Collaboration Skills
- Personalized Medicine and Health Informatics

ACADEMIC QUALIFICATION

- **Ph.D.** (2021), Centre for Interdisciplinary Research in Basic Sciences, JMI University, New Delhi, India.
  Thesis title: *Network Meta-Analysis of Human Gene Expression Profiles to Identify Potential Therapeutic Targets for Tuberculosis Infection.*

- **M.Phil.** (2015), Centre for Interdisciplinary Research in Basic Sciences, JMI University, New Delhi, India.

- **Master of Science** (2013), Bioinformatics, Jamia Millia Islamia University, New Delhi, India

- **Bachelor of Science** (Hon.) (2010), Chemistry, Aligarh Muslim University, India.

SCIENTIFIC SOCIETIES

- Life Member of "Bioinformatics and Drug Discovery Society (BIDDS17-133)", India.

KEY ACHIEVEMENTS (PROJECTS / GRANTS AWARDED)

- "**Young Scientist Fellowship**", Principal Investigator (PI), Ref. No. R.12014/06/2019-HR, Department of Health Research (DHR), Govt. of India (04/2019-10/2021, 2.5 years).
- "**Senior Research Fellow (SRF)**", Indian Council of Medical Research (ICMR), New Delhi, India. Awarded in 2019 for 3 years [Ref. No. ISRM/11(63)/2019].

LIST OF PUBLICATIONS *(in chronological order, starting with the most recent to 2021)*

- **Aftab Alam,** Siddiqui, M.F., Kali, S., Labania, L., Attoub, S., Fernandez-Cabezudo, M.J., & Al-Ramadi, B.K. "Network Analysis and Machine Learning Approach Identifies HSP90-Associated Synthetic Lethal Targets and Drug Responses in Breast Cancer". **npj Precision Oncology (2026);** https://doi.org/10.1038/s41698-026-01738-x

- Kali, S.K., Remya, P.G., **Aftab Alam,** M., Fernandez-Cabezudo, Aydah M., Uday K., BK. Al-Ramadi. "Rethinking PDAC: From Immunological Silence to Therapeutic Opportunity". **Frontiers Immunology (2026);** https://doi.org/10.3389/fimmu.2026.1909191

- Ahmed, M., Haneefa, S.M., **Aftab Alam,** Arafat, K., Kali, S.K., Mohamed, Y.A., Naser, H.T., Idriss, I., Labania, L., Al-Sbiei, A., Attoub, S., Rincon, M., Fernandez-Cabezudo & BK. Al-Ramadi. "Modulation of the immunogenic landscape in colorectal cancer by mitochondrial DNAJC15 protein". **Molecular Biomedicine (2026);** https://doi.org/10.1186/s43556-026-00466-9.

- **Aftab Alam,** M., Fernandez-Cabezudo, Rifat A Hamoudi, BK. Al-Ramadi. "Targeting protein–protein interactions and their modulators to enable new therapeutic strategies for human diseases". **npj Systems Biology and Applications (2026);** https://doi.org/10.1038/s41540-026-00698-z.

- **Aftab Alam**, Mohd F Siddiqui, Rifat A Hamoudi, Uday Kishore, M. Fernandez-Cabezudo, BK. Al-Ramadi. . "A Systems Biology Approach to Unveil Shared Therapeutic Targets and Pathological Pathways Across Major Human Cancers". **Computational and Structural Biotechnology Journal, (2025);** https://doi.org/10.1016/j.csbj.2025.11.061.

- I. Idriss†, Abdelmoneim H Ali†, **Aftab Alam**†, M. Fernandez-Cabezudo, M. Ayyash, BK. Al-Ramadi. "Differential in vitro cytotoxic effects and metabolomic insights into raw and powdered Manuka honey through UPLC-Q-TOF-MS". **Scientific Reports (2024);** DOI: 10.1038/s41598-024-68387-7.

- N. Tamkeen, A. Farooqui, **Aftab Alam**, Najma, S. Tazyeen, MM. Ahmad, N. Ahmad, Romana Ishrat. "Identification of common candidate genes and pathways for Spina Bifida and Wilm's Tumor using an integrative bioinformatics analysis". **Journal of Biomolecular Structure and Dynamics, (2024);** DOI:10.1080/07391102.2023.2199080.

- Rodina A, Xu C, Digwal CS, Joshi S, Patel Y, Santhaseela AR, Bay S, Merugu S, **Alam Aftab**, Yan P, Yang C. "Systems-level analyses of protein-protein interaction network dysfunctions via epichaperomics identify cancer-specific mechanisms of stress adaptation". **Nature Communications (2023);** DOI:10.1038/s41467-023-39241-7.

- Nikhat Imam†**, Aftab Alam**†, Mohd Faizan Siddiqui, Akhtar Veg, Sadik Bay, Md. Jawed Ikbal Khan, and Romana Ishrat. "Network-Medicine Approach for the Identification of Genetic Association of Parathyroid Adenoma with Cardiovascular Disease and Type-2 Diabetes". **Briefings in Functional Genomics (2023);** DOI:10.1093/bfgp/elac054.

- **Aftab Alam**, Hala Abubaker Bagabir, Armiya Sultan, Mohd Faizan Siddiqui, Nikhat Imam, Mustfa F Alkhanani, Ahmad Alsulimani, Shafiul Haque, and Romana Ishrat. "An Integrative Network Approach to Identify Common Genes for the Therapeutics in Tuberculosis and Its Overlapping Non-Communicable Diseases". **Frontiers in Pharmacology (2022);** DOI: 10.3389/fphar.2021.770762.

- Nabeela Majid, Khursheed Siddiqi, **Aftab Alam**, Sadia Malik, Wareesha Ali, Rizwan Hasan Khan. "Cholic acid inhibits amyloid fibrillation: Interplay of protonation and deprotonation". **International Journal of Biological Macromolecules (2022);** DOI:10.1016/j.ijbiomac.2022.09.019

- **Aftab Alam,** Arbaaz Khan, Nikhat Imam, Mohd Faizan Siddiqui, Md. Zubbair Malik and Romana Ishrat. "Design of an Epitope-Based Peptide Vaccine against the Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2): A Vaccine Informatics Approach". **Briefings in Bioinformatics, (2021);** DOI: 10.1093/bib/bbaa340.

- **Aftab Alam,** Nikhat Imam, Mohd Faizan Siddiqui, Md Kaiser Ali, Mohd Murshad Ahmed, and Romana Ishrat. "Human Gene Expression Profiling Identifies Key Therapeutic Targets in Tuberculosis Infection: A Systematic Network Meta-Analysis". **Infection, Genetics and Evolution (2021);** DOI:10.1016/j.meegid.2020.104649.

PUBLICATIONS & CITATION OVERVIEW

Type of Publication	Number of Publications
Articles in International Refereed Journals	More than **42** (As per Google Scholar)
Books/Book Chapters	**1/3**
Number of Citations	More than **770** citations (As per Google Scholar)
h/i10-index	**15/21** (As per Google Scholar)

Full list of publications

Manuscripts Under Preparation / Submitted (2026–2027)

• Idriss, I., Ahmed, M., Al-Sbiei, Ashraf, **Alam, A.,** Abubacker, A., Bashir, G.H., Mohamed, Y.A., Labania, L., Kishore, U., Al-Marzooq, F., Fernandez-Cabezudo, M.Z., & Al-Ramadi, B.K. *"Gut Microbiota-Dependent Immune Remodeling Mediates the Therapeutic Antitumor Activity of Oral Manuka Honey in Experimental Colorectal Cancer"*. **Frontiers Immunology (2026); (Manuscript submitted).**

• Chkier, D.†, **Alam, A.†,** Al-Sbiei, A., Mohamed, Y.A., Fernandez-Cabezudo, M.J., Liu, W., & Al-Ramadi, B.K. "*MyD88 regulates the threshold of antigen receptor signaling in B lymphocytes"*. **(2026); (Manuscript in preparation).**

• **Alam, A.†,** Kali, S.K.† & Al-Ramadi, B.K. "Multi-Omics Profiling Identifies and Validates Prognostic Biomarkers and Therapeutically Relevant Cellular States in Pancreatic Ductal Adenocarcinoma**". (2027); (Manuscript in preparation).**

† These authors contributed equally.

I hereby certify that the information provided in this CV is true, accurate, and complete to the best of my knowledge and belief.



Dr. Aftab Alam